



# Project 2: OpenStreetMap (OSM)

Team Buffalo  
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# Introduction to OpenStreetMap

OpenStreetMap is a collaborative community mapping tool that was created in response to national and private sector restrictions on public access to geodata. Studying OSM involves looking at how data accuracy changes and is distributed in a tool that is run by community members. Certain areas are documented more heavily than others, but there are 1.2 million unique contributors, some of who are actively adding and editing elements on OSM on a daily basis. This project looks at how community participation affects data in OSM and how outside stakeholders use OSM in community relief projects, urban planning, and much more.

# Our Texts and Why We Choose Them

Biljecki, F., Chow, Y. S., & Lee, K. (2023). **Quality of crowdsourced geospatial building information: A global assessment of OpenStreetMap attributes**. Building and Environment, 237, 110295. <https://doi.org/10.1016/j.buildenv.2023.110295>

Vargas-Munoz, J. E., Hu, F., Bastani, F., Wang, X., Delahaye, J., Demir, I., & Lobry, S. (2020). **OpenStreetMap: Challenges and opportunities in machine learning and remote sensing**. IEEE Geoscience and Remote Sensing Magazine, 9(1), 184-199. <https://ieeexplore.ieee.org/abstract/document/9119753>

Zielstra, D., Hochmair, H. H., & Neis, P. (2013). **Assessing the effect of data imports on the completeness of OpenStreetMap – A United States case study**. Transactions in GIS, 17(3), 315-334. <https://doi.org/10.1111/tgis.12037>

Mahmud, Z., Chauhan, A., Sarkar, D., & Soden, R. (2022). **Revisiting engagement in humanitarian mapping: An updated analysis of contributor retention in OpenStreetMap**. In Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems (CHI EA '22), Article 251, 1-6. ACM. <https://doi.org/10.1145/3491101.3519728>

Budhathoki, N. R., Bruce, S., & Nedović-Budić, Z. (2010). **Emergence of the geographic information commons: A case study of OpenStreetMap**. Geographical Review, 100(4), 501-517. [https://icaci.org/files/documents/ICC\\_proceedings/ICC2013/extendedAbstract/401\\_proceeding.pdf](https://icaci.org/files/documents/ICC_proceedings/ICC2013/extendedAbstract/401_proceeding.pdf)

Sehra, S., Singh, J., & Rai, H. (2013). **Assessment of OpenStreetMap data: A review**. International Journal of Computer Applications, 76(1), 10.5120/13331-0888. <https://doi.org/10.5120/13331-0888>

Haklay, M. (2010). **How good is volunteered geographical information? A comparative study of OpenStreetMap and Ordnance Survey datasets**. Environment and Planning B: Planning and Design, 37(4), 682-703. <https://doi.org/10.1068/b35097>

Zielstra, D., & Zipf, A. (2010). **Quantitative studies on the data quality of OpenStreetMap in Germany**. International Journal of Geographical Information Science, 24(11), 1437-1450. [https://www.researchgate.net/publication/267989860\\_Quantitative\\_Studies\\_on\\_the\\_Data\\_Quality\\_of\\_OpenStreetMap\\_in\\_Germany](https://www.researchgate.net/publication/267989860_Quantitative_Studies_on_the_Data_Quality_of_OpenStreetMap_in_Germany)

OpenStreetMap Wiki. (n.d.). **Beginners' guide to OSM**. OpenStreetMap Wiki. [https://wiki.openstreetmap.org/wiki/Beginners%27\\_Guide](https://wiki.openstreetmap.org/wiki/Beginners%27_Guide)

# Final Social Worlds Diagram

## Stakeholders:

### Macro

National and continental development:  
Widely used among people in the area of governance for research purposes  
Major Corporations such as Apple, Google, FourSquare, Flickr  
Google runs a competing map database that does not have free license for commercial use  
Private sector entities utilizing geographical information

### Meso

Built environment  
Architectural and precinct development  
Researchers that use OSM data to study diseases, urban planning, energy consumption in different areas  
Transportation planners  
Urban developers

### Micro

Local citizens using maps that use OSM data

## Data Sources:

### Macro

OSM contains geographical data on over 100 million buildings worldwide and is available with a liberal license  
Data sets provided from governments, where the data source allows, it does not conflict with existing OSM data and the quality is great  
1.2 million contributors to the platform  
Public Domain Data collected by governments  
Public domain data that is updated periodically  
A US census, TIGER/Line Import was OSM generated a Foundation for the street map data available in OSM

### Meso

Companies collecting data  
Academia often collect, maintain and release building data openly  
Some machine learning is being used to improve poor user generated OSM data  
Geographic information systems  
Remote sensing technologies  
Using existing OSM layers to train models  
GPS enabled Devices can give data to volunteers to port into OSM

### Micro

Building attributes such as the type and height feed many studies  
Data on buildings taken as they are more detailed with more geographical detail information and details about number of attributes  
Smaller amounts of data found from volunteers  
Annotations on the individual elements in OSM made by community members

## Influences:

### Macro

General amounts of building data are increasing and allow for data to become higher quality  
Changes made to the licensing model by Google Maps in early 2012, which limits the usage of free Map APIs  
This motivated companies to switch from Google Maps to OSM for their application's map data

### Meso

Originally was focused on streets, but grew to various other points of interest  
Used in studies about urban planning and microclimates  
Area type influences quality of data

### Micro

Quality of the data collected and put into OSM depends on the volunteer and their commitment

# Stakeholders:

## Macro

National and continental development:

Widely used among people in the area of geoscience for research purposes

Major Corporations such as: Apple, Craigslist, FourSquare, Flickr

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Built environment

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Researchers that use OSM data to study disasters, urban planning, energy consumption in different areas

Transportation planners

Urban developers

## Micro

Local citizens using maps that use OSM data

# Data Sources:

## Macro

OSM contains geospatial data on over 500 million buildings worldwide and is available with a liberal license

Data sets provided from governments - only when the data license allows, it does not conflict with existing OSM data, and the quality is good

1.2 million contributors to the platform

Public Domain Data collected by governments

Public domain data that is updated periodically

A US census TIGER/Line import into OSM generated a foundation for the road map data available in OSM

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Geographic information systems

Remote sensing technologies

Using existing OSM layers to train models

GPS enabled Devices can give data to volunteers to port into OSM

## Micro

Building attributes such as the type and height feed many studies

Data on buildings in urban areas are more detailed - with more geographically detailed information and details about number of stories/floors

Smaller amounts of data found from volunteers

Annotations on the individual elements in OSM made by community members

# Influences:

## Macro

General amounts of building data are increasing and allow for data to become higher quality

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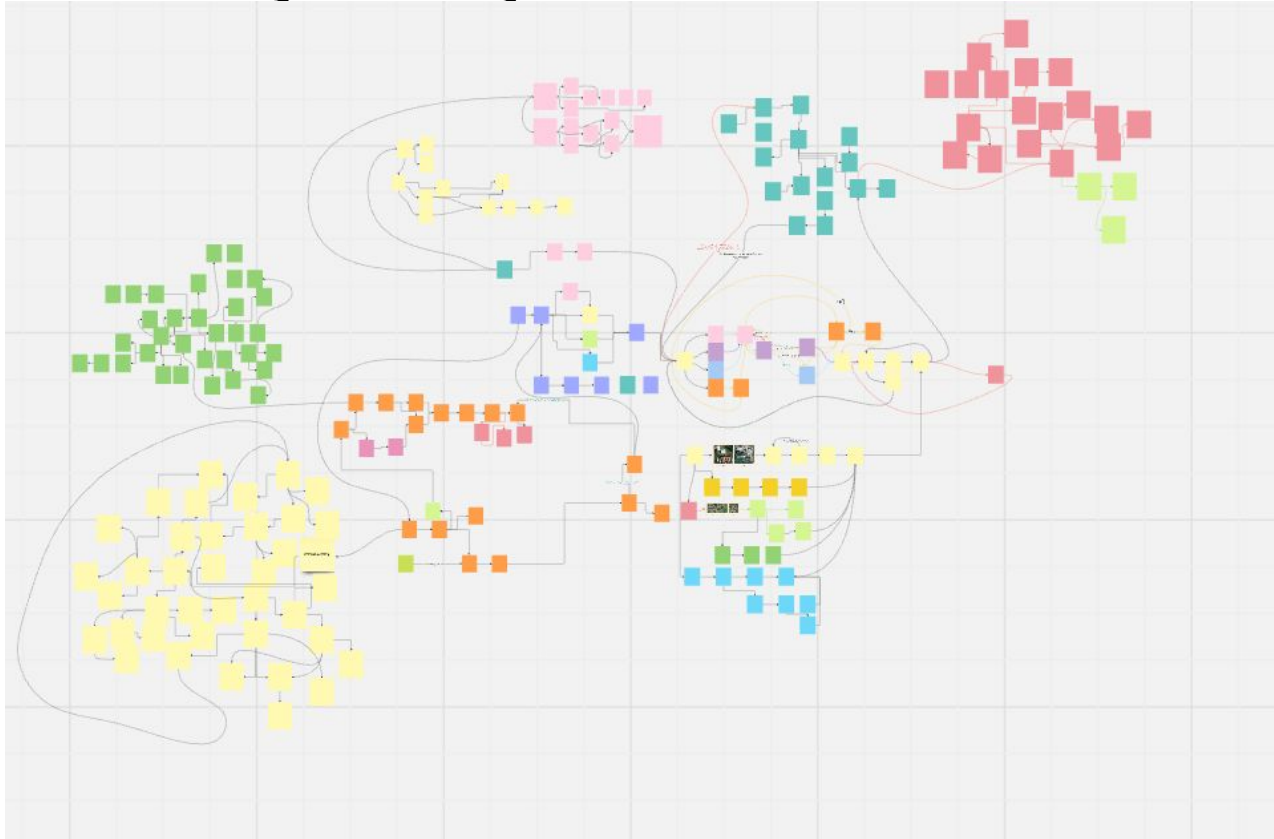
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# The Final Data Journey

<https://miro.com/app/board/uXjVLXEd40Q=/>





# Frictions and Mutabilities

- **Lack of user contribution**
  - There are many barriers for contributing to volunteered geographic information (VGI) projects, including but not limited to: lack of suitable equipment, lack of time or a shift of priorities, users don't feel the need, or other technical barriers. Upon further analysis, it was found that many users were familiar with OSM but don't actively contribute. Only about 30% of the registered users have ever contributed or altered data.
  - With such a large number of users not able to access or understand the tools required to add elements to OSM, many users stop contributing immediately. This creates gaps in information in more rural areas where these tools are not as accessible.
- **User mapping experience and accuracy**
  - Crowdsourcing allows for the collection of huge data with the help of general public, however, the issue lies with varying levels of user mapping experience. Because of this, we find that not all data contributions are equal or of the same quality. There are often inconsistencies and inaccuracies found in data contributions from users who don't fully understand the platform or how to navigate it.
  - This inaccuracy in data can affect how researchers, humanitarian organizations, and the general public use tools that involve OSM. If the data are inaccurate, these people and communities can make decisions that they would otherwise not make if the data was correct.
- **Keeping existing data up-to-date**
  - As new aerial satellite data is imported to OSM on a large scale, it can affect the accuracy of the elements that are already in OSM. This creates a large amount of user editing that need to be done to correct the errors created by new imagery.
  - This temporary inaccuracy in information takes a lot of time and effort to fix, taking away time that could be spent adding new elements and nodes into OSM.

Biljecki, Filip, Yoong Shin Chow, and Kay Lee. "Quality of crowdsourced geospatial building information: A global assessment of OpenStreetMap attributes." *Building and Environment* 237 (2023): 110295.

[https://www.sciencedirect.com/science/article/pii/S0360132323003220?ref=pdf\\_download&fr=RR-2&rr=8cf7bfb6e8f61f35](https://www.sciencedirect.com/science/article/pii/S0360132323003220?ref=pdf_download&fr=RR-2&rr=8cf7bfb6e8f61f35)

#### **Stakeholders:**

##### **Macro:**

- national and continental development

##### **Meso:**

- built environment
- architectural and precinct development
- Researchers that use OSM data to study disasters, urban planning, energy consumption in different areas

##### **Micro:**

- local citizens using maps that use OSM data

#### **Data Sources:**

##### **Macro:**

- OSM contains geospatial data on over 500 million buildings worldwide and is available with a liberal license
- Data sets provided from governments - only when the data license allows, it does not conflict with existing OSM data, and the quality is good
- 1.2 million contributors to the platform

##### **Meso:**

- Companies collecting data
- Academia often collect, maintain and release building data openly

##### **Micro:**

- building attributes such as the type and height feed many studies
- Data on buildings in urban areas are more detailed - with more geographically detailed information and details about number of stories/floors
- Smaller amounts of data found from volunteers

#### **Influences:**

##### **Macro:**

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##### **Meso:**

- Originally was focused on streets, but grew to various other points of interest
- Used in studies about urban planning and microclimates
- Area type influences quality of data

##### **Micro:**

- Quality of the data collected and put into OSM depends on the volunteer and their commitment

Vargas-Munoz, John E., et al. "OpenStreetMap: Challenges and opportunities in machine learning and remote sensing." *IEEE Geoscience and Remote Sensing Magazine* 9.1 (2020): 184-199.

<https://ieeexplore-ieee-org.colorado.idm.oclc.org/stamp/stamp.jsp?tp=&arnumber=9119753&tag=1>

#### **Stakeholders:**

##### **Macro:**

- Widely used among people in the area of geoscience for research purposes

##### **Meso:**

- built environment
- architectural and precinct development
- Researchers that use OSM data to study disasters, urban planning, energy consumption in different areas

##### **Micro:**

- local citizens using maps that use OSM data

#### **Data Sources:**

##### **Macro:**

- Public Domain Data collected by governments

##### **Meso:**

- Some machine learning is being used to improve poor user generated OSM data
- geographic information systems
- remote sensing technologies
- using existing OSM layers to train models

##### **Micro:**

- annotations on the individual elements in OSM made by community members

#### **Influences:**

##### **Macro:**

- OSM was created as an alternative to authoritative sources of geospatial data that was not made accessible to many groups

##### **Meso:**

- In rural areas annotations suffer from 3 problems:
  - they are geometrically misaligned
  - some annotations do not correspond to buildings in the updated aerial imagery
  - some OSM annotations are missing for buildings that are present in the updated aerial imagery

##### **Micro**

Zielstra, Dennis, Hartwig H. Hochmair, and Pascal Neis. "Assessing the effect of data imports on the completeness of openstreetmap-a united states case study." *Transactions in GIS* 17.3 (2013): 315-334.

<https://onlinelibrary.wiley.com/doi/full/10.1111/tgis.12037>

#### **Stakeholders:**

##### **Macro:**

- Major Corporations such as: Apple, Craigslist, FourSquare, Flickr
- Google runs a competing map database that does not have free license for commercial use

##### **Meso:**

- Users of these companies products

##### **Micro:**

#### **Data Sources:**

##### **Macro:**

- Public domain data that is updated periodically
- A US census TIGER/Line import into OSM generated a foundation for the road map data available in OSM

##### **Meso:**

- GPS enabled Devices can give data to volunteers to port into OSM

##### **Micro:**

- Contributors

#### **Influences:**

##### **Macro:**

- changes made to the licensing model by Google Maps in early 2012, which limits the usage of free Map APIs
- This motivated companies to switch from Google Maps to OSM for their application's map data

##### **Meso:**

##### **Micro:**

Zarif Mahmud, Aarjav Chauhan, Dipto Sarkar, and Robert Soden. 2022. **Revisiting Engagement in Humanitarian Mapping: An Updated Analysis of Contributor Retention in OpenStreetMap**. In Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 251, 1–6.

<https://dl.acm.org/doi/abs/10.1145/3491101.3519728>

*summary here*

Stakeholders: 1mil+ unique user accounts, researchers, people around the world, development agencies,

Data Sources: crowdsourcing, metadata, global contributors, OSM, open-data, mapathons

Outside Influences: existing roads and houses, studies, open software projects, contributor retention/engagement

Budhathoki, N. R., Bruce, S., & Nedović-Budić, Z. (2010). **"Emergence of the Geographic Information Commons: A Case Study of OpenStreetMap."** *Geographical Review*, 100(4), 501-517.

[https://icaci.org/files/documents/ICC\\_proceedings/ICC2013/extendedAbstract/401\\_proceeding.pdf](https://icaci.org/files/documents/ICC_proceedings/ICC2013/extendedAbstract/401_proceeding.pdf)

*this is a case study looking at collaborative data collection and creation, it talks about the importance of community engagement of OSM as a valuable "information commons"/public geographic resource.*

Stakeholders: inactive/active OSM contributors, academic institutions

Data Sources: survey for 516 participants, OSM platforms/surveys/userbase

Outside Influences: time constraint (for non-contribution), lack of gps services,

Sehra, Sukhjot & Singh, Jaiteg & Rai, Hardeep. (2013). **Assessment of OpenStreetMap Data - A Review**.

International Journal of Computer Applications. 76. 10.5120/13331-0888.

[https://www.researchgate.net/publication/257028692\\_Assessment\\_of\\_OpenStreetMap\\_Data\\_-\\_A\\_Review](https://www.researchgate.net/publication/257028692_Assessment_of_OpenStreetMap_Data_-_A_Review)

*OSM and crowd sourcing !*

Stakeholders: general public who use maps, researchers,

Data Sources: Volunteered Geographic Information, contribution by the

end users, crowdsourcing, proprietary data or data of governmental map agencies, collected data with help of general public with varying level of mapping experience

Outside Influences: crowdsourcing models, technological advancements, gov restrictions



**Haklay, M. (2010). How Good is Volunteered Geographical Information? A Comparative Study of OpenStreetMap and Ordnance Survey Datasets. *Environment and Planning B: Planning and Design*, 37(4), 682-703.**

**Stakeholders:** general public, government agencies, volunteers contributing to OpenStreetMap (OSM), and private sector entities utilizing geographical information.

**Data Sources:** OSM data, Ordnance Survey (official geographical data from the UK), volunteered geographical information (VGI).

**Outside Influences:** Availability of official government data sources, technological advancements in data collection, the growing role of crowdsourcing in geographical data, and the increasing reliance on open data platforms by both public and private sectors.

**Zielstra, D., & Zipf, A. (2010). Quantitative studies on the data quality of OpenStreetMap in Germany. *In International Journal of Geographical Information Science*.**

**Stakeholders:** German government, transportation planners, urban developers, OSM contributors, and academic researchers.

**Data Sources:** OSM datasets, comparative studies of traditional GIS data sources in Germany, and user-generated contributions to OSM.

**Outside Influences:** National and regional mapping authorities in Germany, academic studies on spatial data accuracy, and the evolving role of open-source geospatial data in public and private projects.

**OpenStreetMap Wiki – Beginners' Guide to OSM.**

**Stakeholders:** Volunteers and new users contributing to OSM, community-driven mappers, and organizations relying on OSM data.

**Data Sources:** User-contributed map data, public datasets that are integrated into OSM, and GPS device data from volunteers.

**Outside Influences:** Policies around open data and map contributions, technological developments in mapping software and tools, and the increasing global demand for up-to-date geographical information.